

# Super-Eddington Little Blue Dots Reionize Helium Too Early

JWST's faint AGN may over-ionize helium, forcing super-Eddington fractions below 10%.

arXiv:2608.28745 — Super-Eddington LBDs and helium reionization timing.

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## Background

JWST found many faint broad-line AGN at  $z > 4$ , called Little Blue Dots. Some models link them to super-Eddington accretion with strong extreme-UV and weak X-rays.

Does this super-Eddington population reionize helium earlier than observed?

## What this paper is selling

The authors compute helium reionization histories for super-Eddington spectra. They show that helium timing constraints limit the super-Eddington fraction to  $\lesssim 10\%$ .

## Key results to remember

- Helium reionization ending after  $z \approx 3$  disfavors mostly super-Eddington LBDs.
- Super-Eddington fraction must be  $\lesssim 10\%$  for fiducial abundance and escape.
- Majority super-Eddington requires escape fractions  $\lesssim 1.5\%$  or lower abundance.

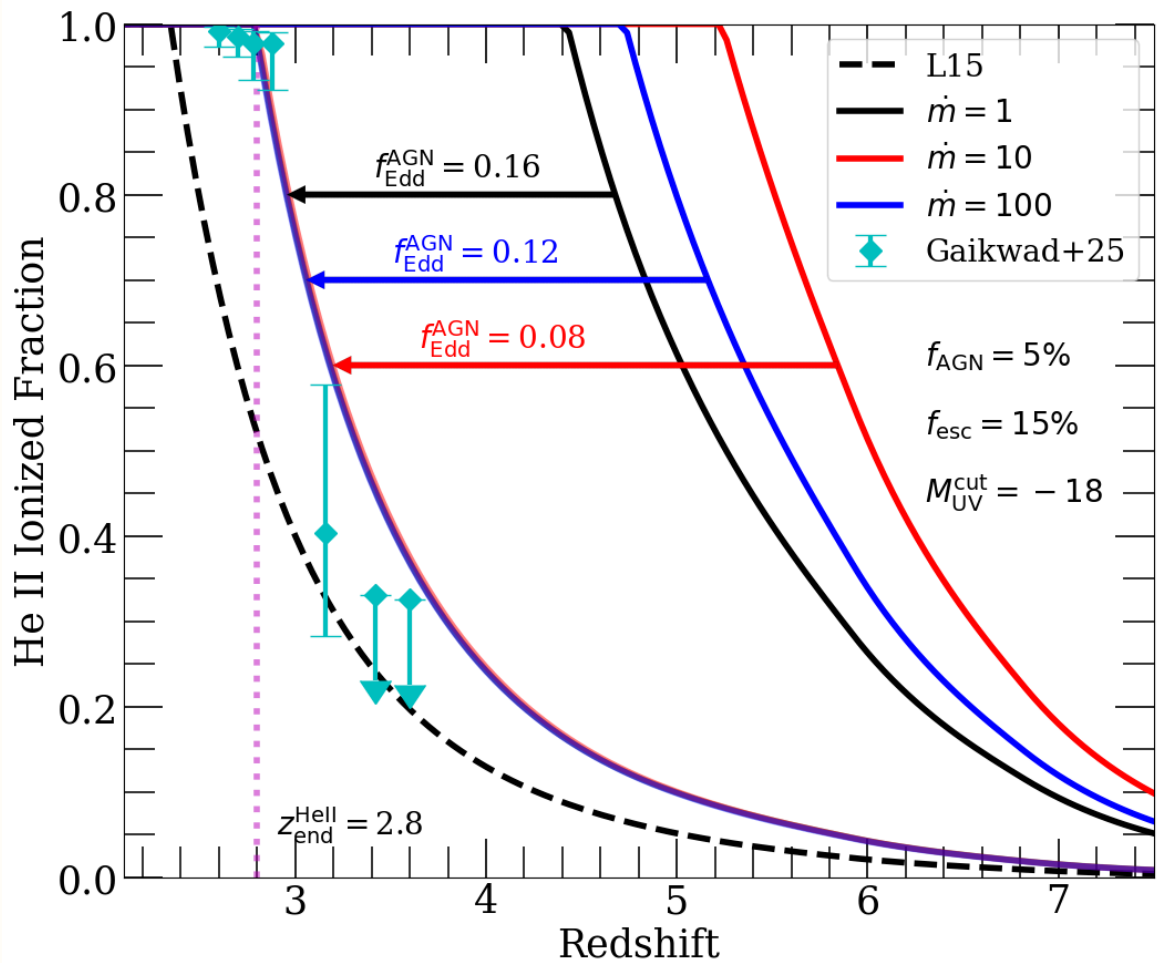


Figure 3 | Helium reionization histories for different SEDs, showing required super-Eddington fractions to match  $z = 2.8$  completion.

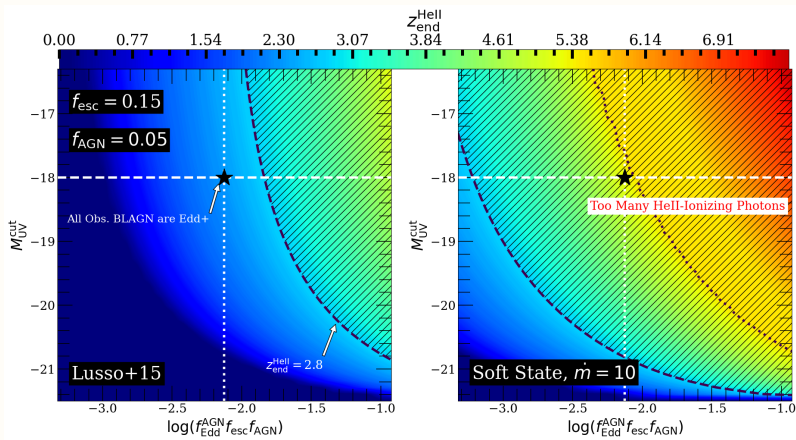


Figure 4 | Excluded AGN parameter space from helium reionization constraints, with hatched regions disfavored by data.